Do Now · retrieval
Iteration in C#

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the difference between = and == in C#?

Q2

Give one reason Unicode may be chosen instead of standard ASCII.

Q3

Convert 01011000 0011 to denary using an 8-bit two’s-complement mantissa and a 4-bit two’s-complement exponent.

Q4

mark=85. The selection is: if mark >= 80 then print("Distinction"); else if mark >= 50 then print("Pass"); else print("Fail"). Determine the output and explain which branch runs.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Iteration repeats instructions — Iteration means executing a block of instructions repeatedly. The loop needs a rule that determines whether another repetition happens.

Core idea 2

for = count-controlled — CODE:
for (int i = 0; i < 5; i++)
{
Console.WriteLine(i);
} — The header sets the counter, tests whether the loop continues, then changes the counter after each repetition.

Core idea 3

while = test before repeating — A while loop is useful when you do not know how many attempts will be needed. — CODE:
while (valid == false)
{
// ask again
}

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

do...while = run first, test after — CODE:
do
{
Console.Write("Password: ");
answer = Console.ReadLine();
}
while (answer != "computer"); — The body executes before the condition is checked, so it always runs at least once.

Vocabulary · use the correct technical terms

Iteration — Repeating a block of instructions.

Count-controlled iteration — Repetition where the number of repeats is known or controlled by a counter.

Condition-controlled iteration — Repetition that continues according to a Boolean test.

Counter — A variable used to track the number of repetitions.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Build a times-table program

Follow the worked method

1. Read the number — CODE:
Console.Write("Number: ");
int number =
int.Parse(Console.ReadLine()); — This happens once before the repetition begins.

2. Calculate each multiple — CODE:
for (int i = 1; i <= 5; i++)
{
int answer = number * i;
Console.WriteLine(answer);
} — The same two instructions run five times, but i changes each time.

3. Trace a known input — CODE:
number = 7

i = 1 → 7
i = 2 → 14
i = 3 → 21
i = 4 → 28
i = 5 → 35

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Complete the pattern
CODE:

```
for (int i = ?; i <= ?; i++)
{
Console.WriteLine(?);
}
```

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Countdown
Use a for loop to output:
10, 9, 8, ... 1

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Password attempts
Use do...while to keep asking until the password is "computer" .
Count how many attempts were made and output the total after the correct password.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A while loop always executes at least once.

Use key vocabulary from memory

1. Explain Iteration in your own words.

2. Explain Count-controlled iteration in your own words.

3. Explain Condition-controlled iteration in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.